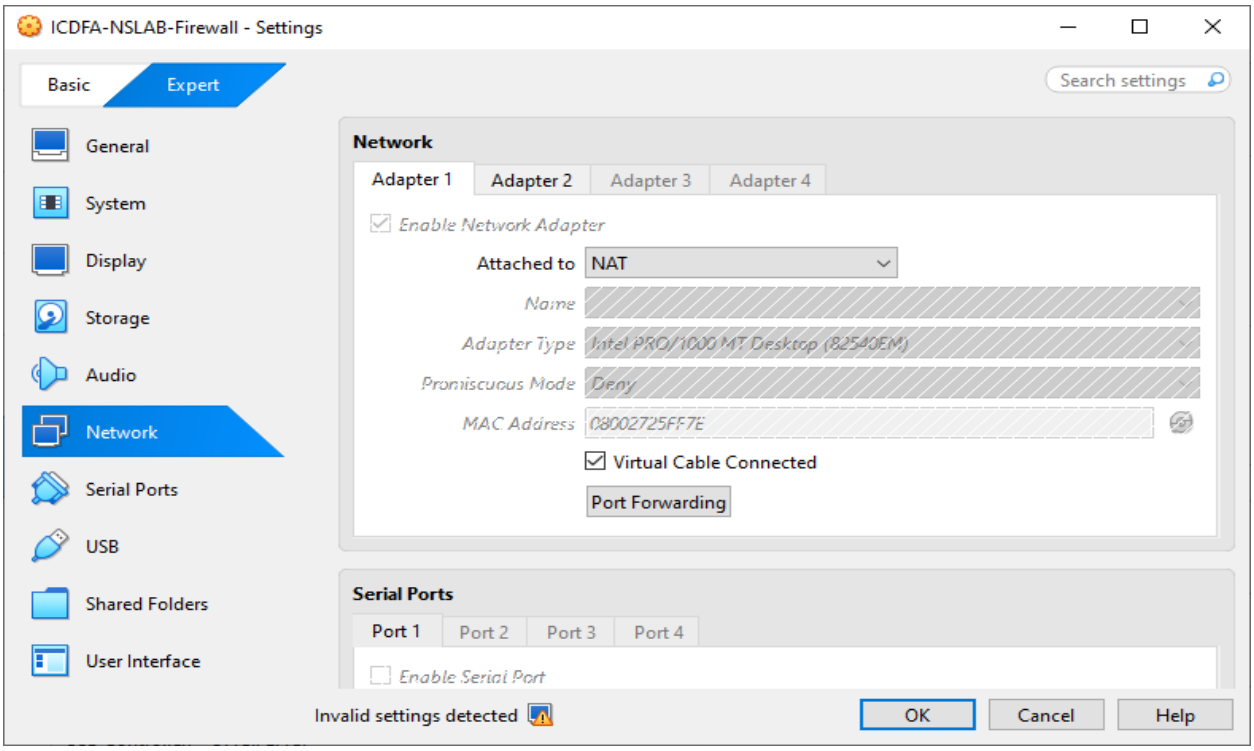
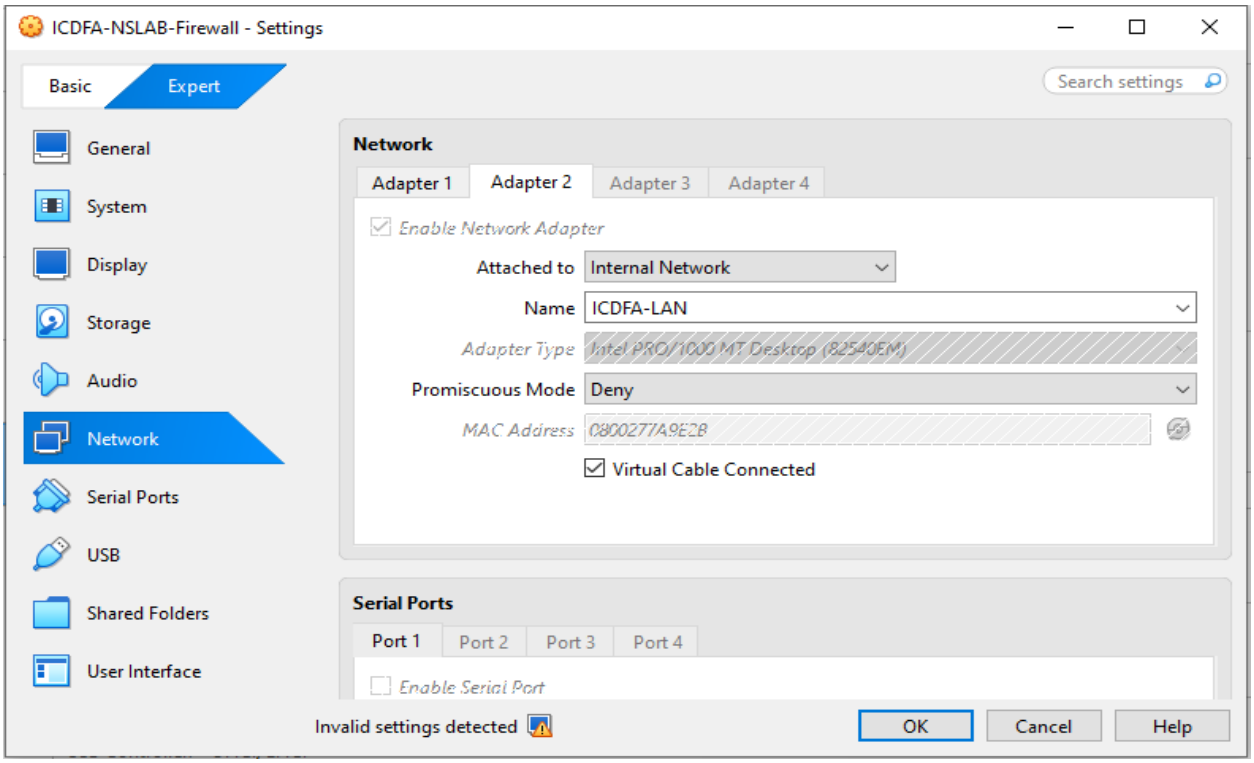
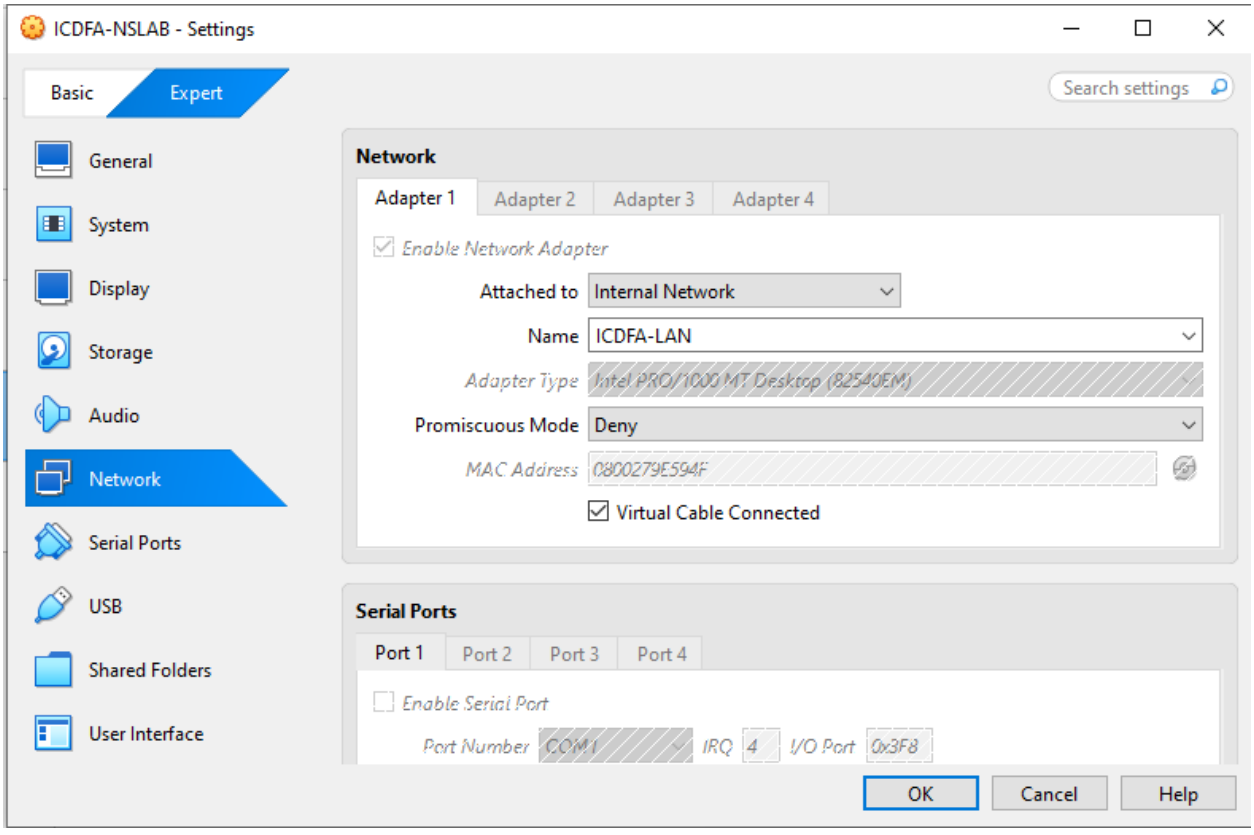
Full name: Adeyemo Adetoye
Registration number: C11/26/FCDF/17140
Linux distribution / version: Version 7.2.10r174163
VM hostname: Kali




ICDFA-NSLAB-Firewall - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Internal Network
Name
ICDFA-LAN
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
0800277A9E2B
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
Invalid settings detected
OK
Cancel
Help


ICDFA-NSLAB - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Internal Network
Name
ICDFA-LAN
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
0800279E594F
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
Port Number
COM1
IRQ
4
I/O Port
0x3F8
OK
Cancel
Help

```
Starting Cron: OK


>>> Invoking start script 'openvpn'
>>> Invoking start script 'sysctl'
Service `sysctl' has been restarted.
>>> Invoking start script 'beep'
Root file system: zroot/ROOT/default
Tue Oct  6 16:13:00 WAT 2026

*** icdfa-nslab-firewall-v1.icdfa.test: OPNsense 26.7 (amd64) ***

 LAN (em1)       -> v4: 10.10.10.1/24
 WAN (em0)       -> v4/DHCP4: 10.0.2.15/24
                    v6/DHCP6: fd17:625c:f037:2:a00:27ff:fe25:ff7e/64

 HTTPS: SHA256 E8 0B 30 3E D5 89 EB 63 A0 D8 83 2F E6 38 4C 22
               82 2C 5A A4 39 D5 45 54 57 50 A0 73 58 E0 46 90
 SSH:   SHA256 K1OLtdmW484dUmuPy37tD714RAODe5LKjSU/+bKB1Tw (ECDSA)
 SSH:   SHA256 5wvyP8AD885sP0fHz7CgfNStFMhPwdXX42x23cBxco4 (ED25519)
 SSH:   SHA256 6e/0XrjSdfL/Hsy8atZw0IPQGsb1P6zFfuj/FZ5Od+A (RSA)

FreeBSD/amd64 (icdfa-nslab-firewall-v1.icdfa.test) (ttyv0)

login:
```

```
|  Code:        https://github.com/opnsense    |          `:::::::::
|  Reddit:      https://reddit.com/r/opnsense  |           `:::::::
-----------------------------------------------

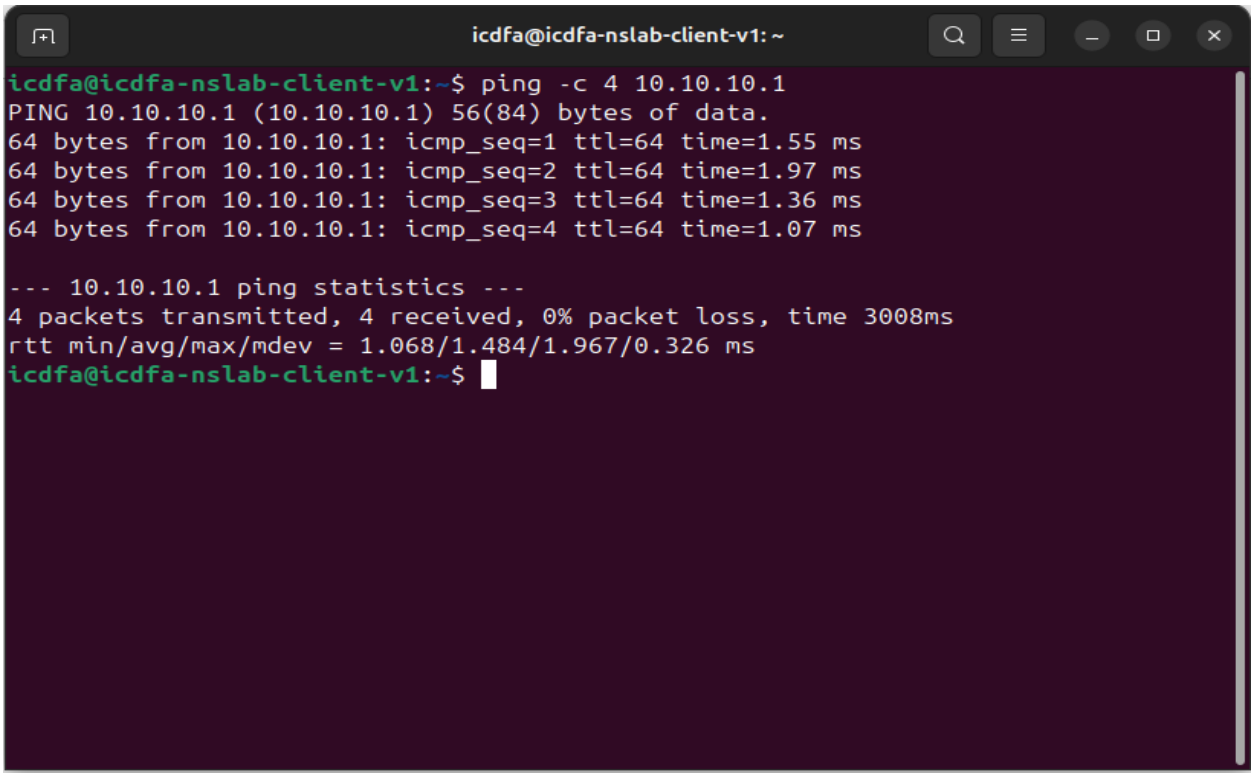
*** icdfa-nslab-firewall-v1.icdfa.test: OPNsense 26.7 (amd64) ***

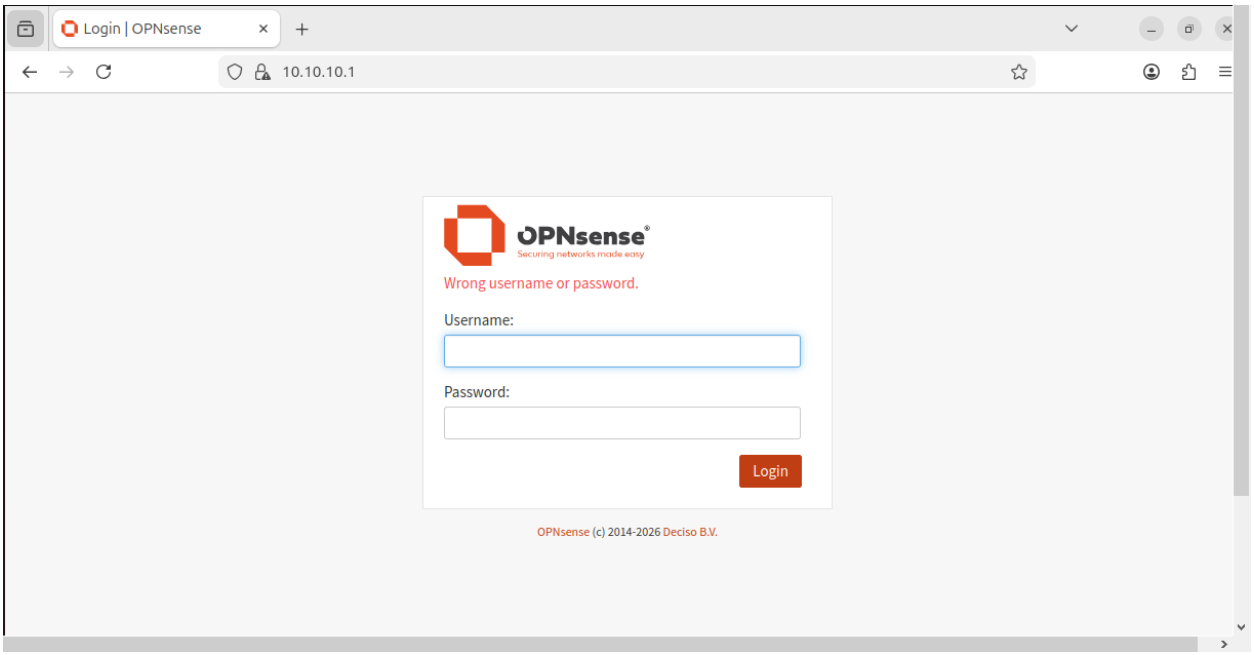
 LAN (em1)       -> v4: 10.10.10.1/24
 WAN (em0)       -> v4/DHCP4: 10.0.2.15/24
                    v6/DHCP6: fd17:625c:f037:2:a00:27ff:fe25:ff7e/64

 HTTPS: SHA256 E8 0B 30 3E D5 89 EB 63 A0 D8 83 2F E6 38 4C 22
               82 2C 5A A4 39 D5 45 54 57 50 A0 73 58 E0 46 90
 SSH:   SHA256 K1OLtdmW484dUmuPy37tD714RAODe5LKjSU/+bKB1Tw (ECDSA)
 SSH:   SHA256 5wvyP8AD885sP0fHz7CgfNStFMhPwdXX42x23cBxco4 (ED25519)
 SSH:   SHA256 6e/0XrjSdfL/Hsy8atZw0IPQGsb1P6zFfuj/FZ5Od+A (RSA)

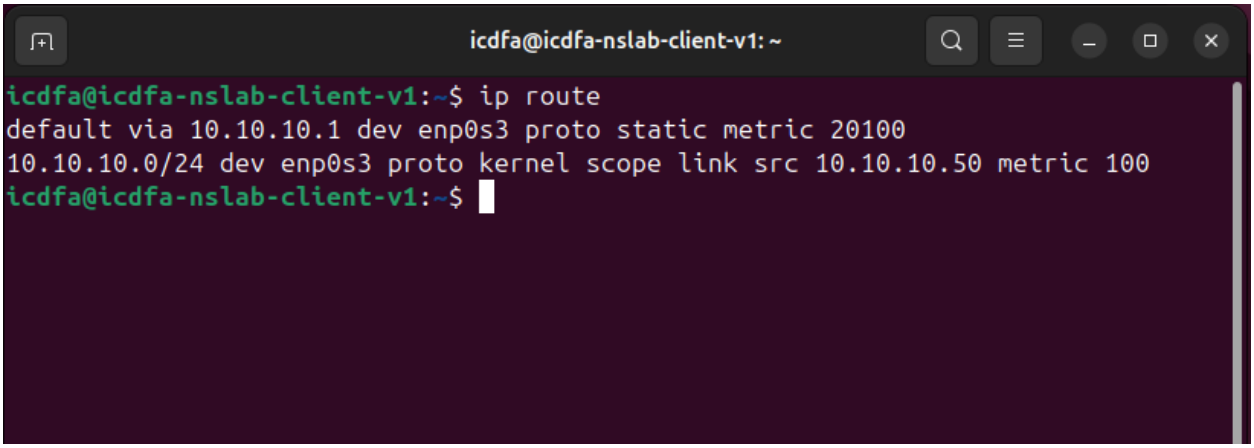
  0) Logout                              7) Ping host
  1) Assign interfaces                   8) Shell
  2) Set interface IP address            9) pfTop
  3) Reset the root password            10) Firewall log
  4) Reset to factory defaults          11) Reload all services
  5) Power off system                   12) Update from console
  6) Reboot system                      13) Restore a backup

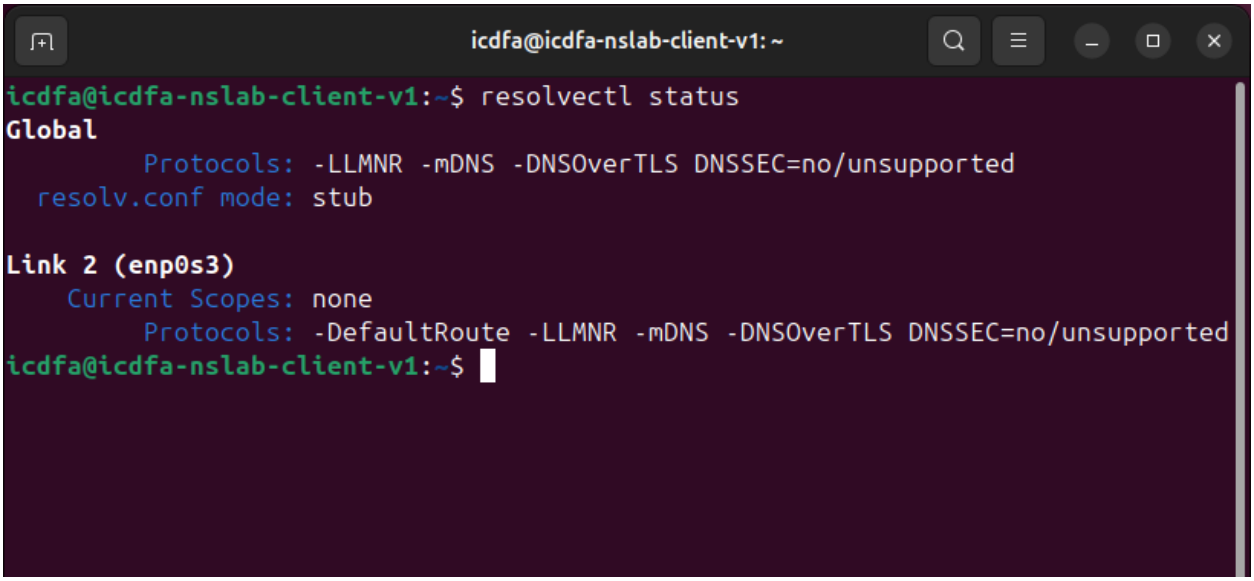
Enter an option:
```

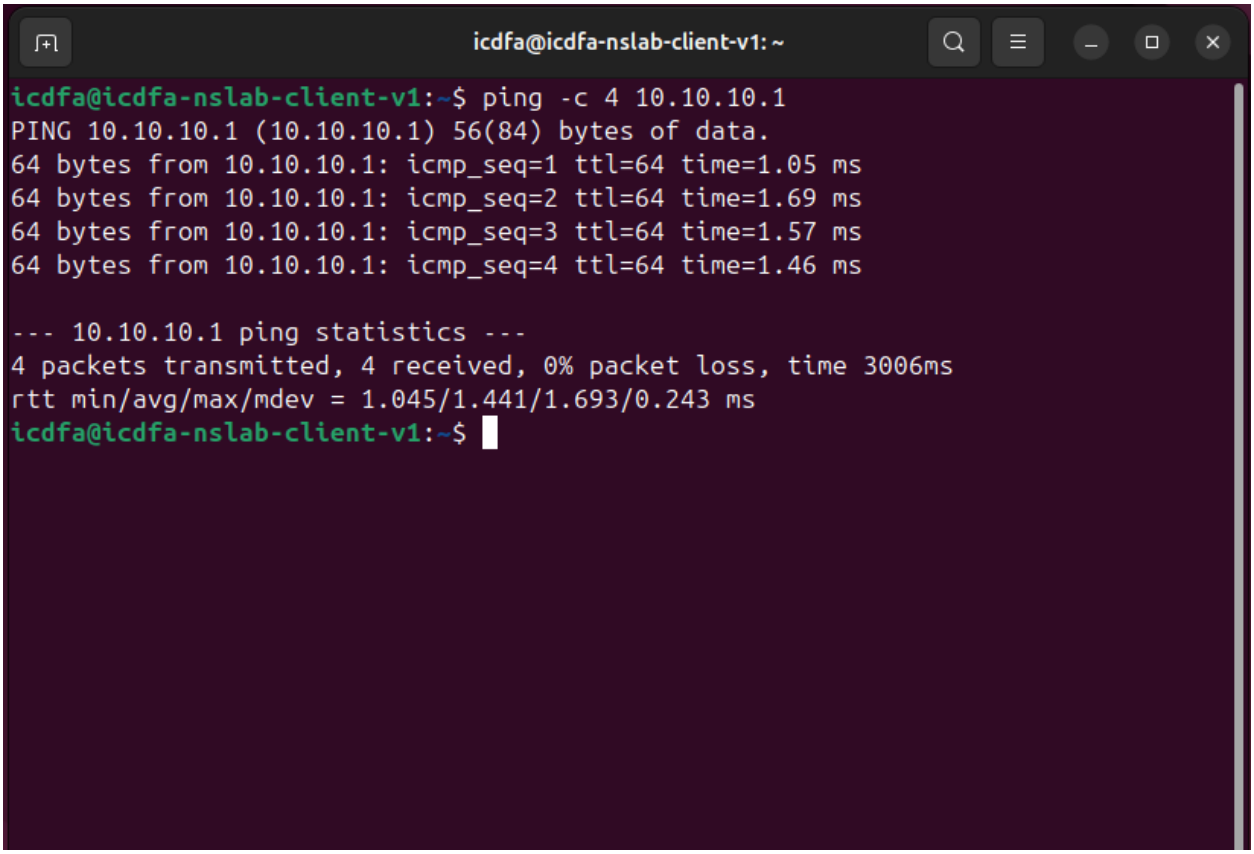

icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=1.55 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=1.97 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=1.36 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=1.07 ms
--- 10.10.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 1.068/1.484/1.967/0.326 ms
icdfa@icdfa-nslab-client-v1:~$

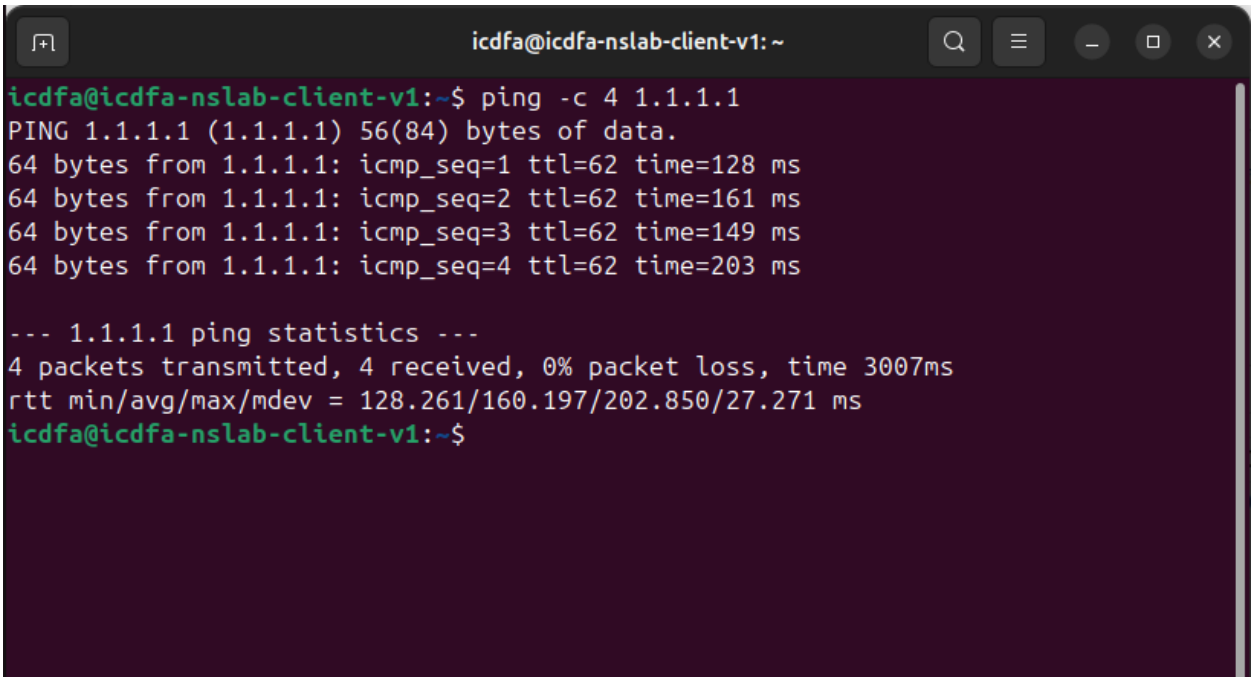

Login | OPNsense
10.10.10.1
OPNsense
Securing networks made easy
Wrong username or password.
Username:
Password:
Login
OPNsense (c) 2014-2026 Deciso B.V.


icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ ip -4 -br address
lo   UNKNOWN   127.0.0.1/8
enp0s3   UP   10.10.10.50/24
icdfa@icdfa-nslab-client-v1:~$

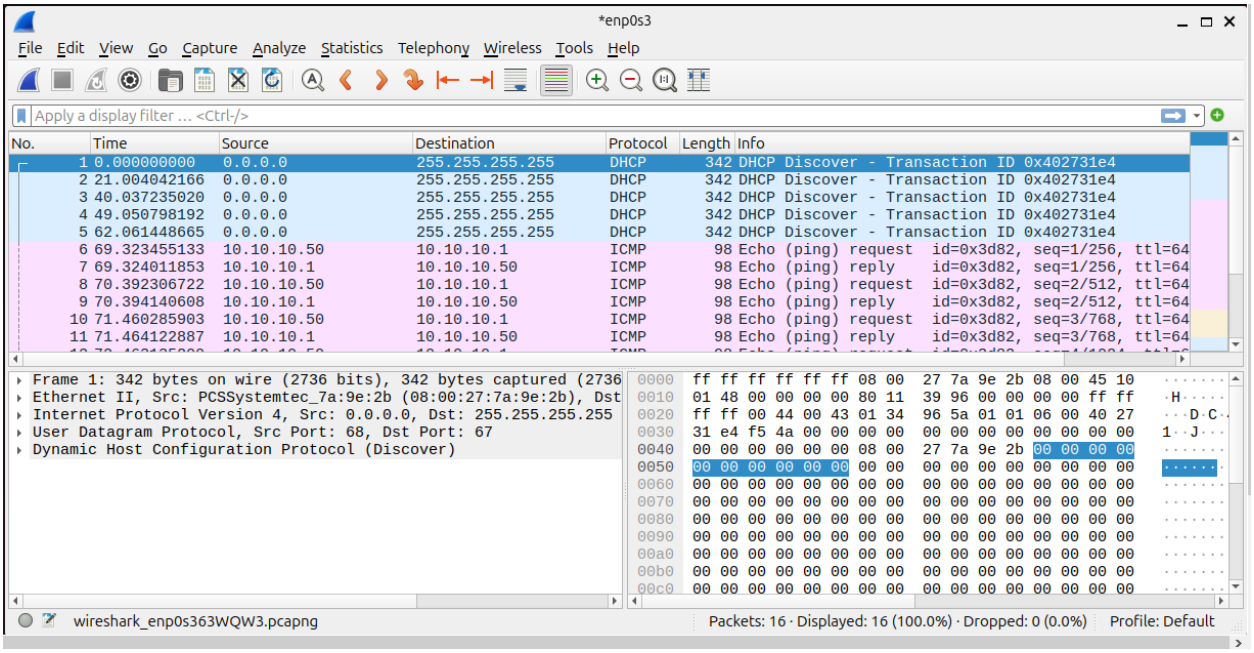

icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ ip route
default via 10.10.10.1 dev enp0s3 proto static metric 20100
10.10.10.0/24 dev enp0s3 proto kernel scope link src 10.10.10.50 metric 100
icdfa@icdfa-nslab-client-v1:~$

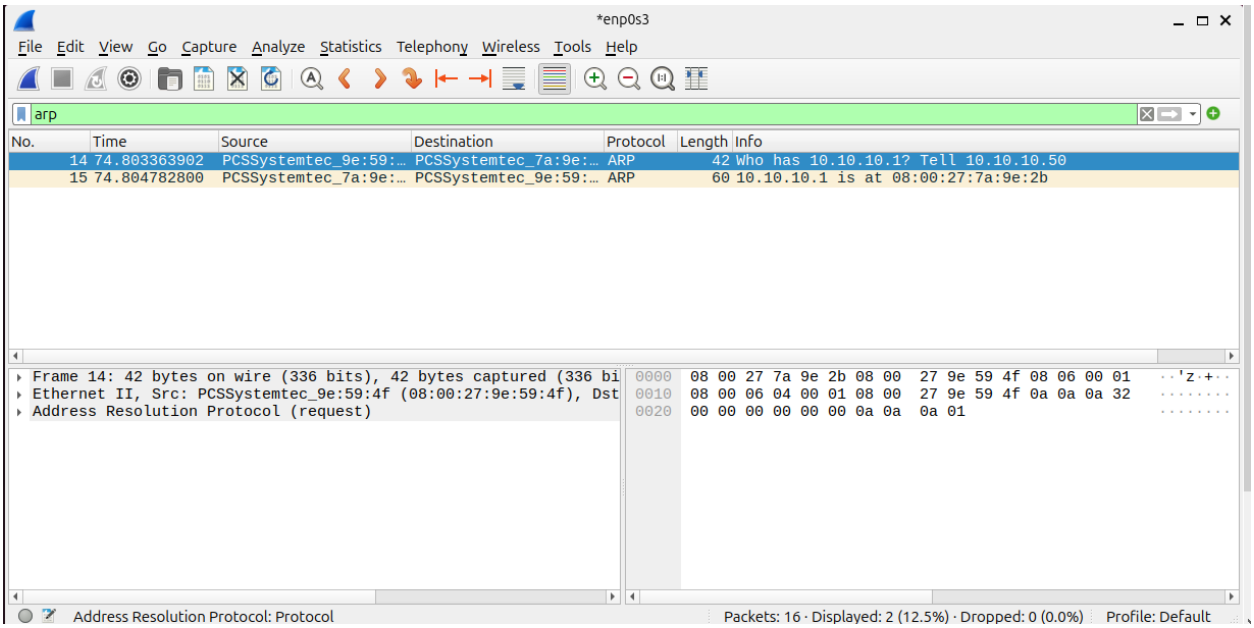

icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ resolvectl status
Global
Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
resolv.conf mode: stub
Link 2 (enp0s3)
Current Scopes: none
Protocols: -DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
icdfa@icdfa-nslab-client-v1:~$

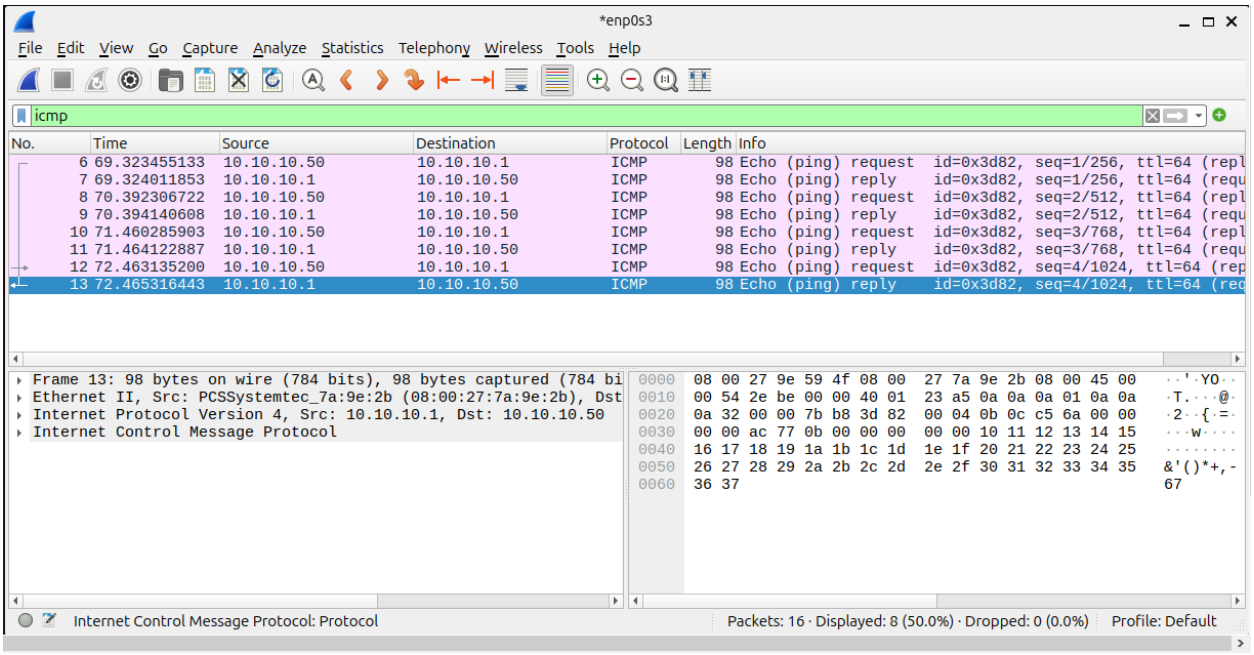

icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=1.05 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=1.69 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=1.57 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=1.46 ms
--- 10.10.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.045/1.441/1.693/0.243 ms
icdfa@icdfa-nslab-client-v1:~$

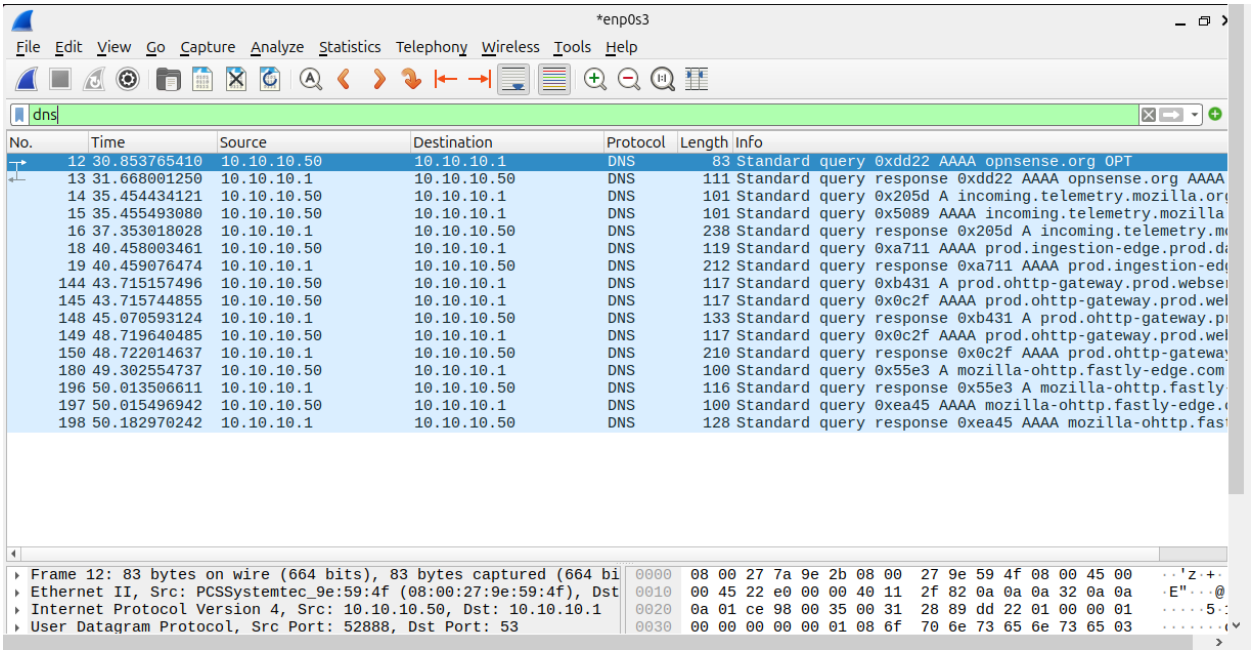

icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=62 time=128 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=62 time=161 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=62 time=149 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=62 time=203 ms
--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 128.261/160.197/202.850/27.271 ms
icdfa@icdfa-nslab-client-v1:~$


icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ getent hosts opnsense.org
2001:1af8:2050:a001:1::1 opnsense.org
icdfa@icdfa-nslab-client-v1:~$ curl -I https://opnsense.org
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 14:44:51 GMT
Server: Apache
Link: <https://opnsense.org/wp-json/>; rel="https://api.w.org/", <https://opnsense.org/wp-json
ternate"; title="JSON"; type="application/json", <https://opnsense.org/>; rel=shortlink
Strict-Transport-Security: max-age=15552000; includeSubDomains; preload
Content-Type: text/html; charset=UTF-8
icdfa@icdfa-nslab-client-v1:~$


*enp0s3
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter … <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 0.0.0.0 255.255.255.255 DHCP 342 DHCP Discover - Transaction ID 0x402731e4
2 21.004042166 0.0.0.0 255.255.255.255 DHCP 342 DHCP Discover - Transaction ID 0x402731e4
3 40.037235020 0.0.0.0 255.255.255.255 DHCP 342 DHCP Discover - Transaction ID 0x402731e4
4 49.050798192 0.0.0.0 255.255.255.255 DHCP 342 DHCP Discover - Transaction ID 0x402731e4
5 62.061448665 0.0.0.0 255.255.255.255 DHCP 342 DHCP Discover - Transaction ID 0x402731e4
6 69.323455133 10.10.10.50 10.10.10.1 ICMP 98 Echo (ping) request id=0x3d82, seq=1/256, ttl=64
7 69.324011853 10.10.10.1 10.10.10.50 ICMP 98 Echo (ping) reply id=0x3d82, seq=1/256, ttl=64
8 70.392306722 10.10.10.50 10.10.10.1 ICMP 98 Echo (ping) request id=0x3d82, seq=2/512, ttl=64
9 70.394140608 10.10.10.1 10.10.10.50 ICMP 98 Echo (ping) reply id=0x3d82, seq=2/512, ttl=64
10 71.460285903 10.10.10.50 10.10.10.1 ICMP 98 Echo (ping) request id=0x3d82, seq=3/768, ttl=64
11 71.464122887 10.10.10.1 10.10.10.50 ICMP 98 Echo (ping) reply id=0x3d82, seq=3/768, ttl=64
Frame 1: 342 bytes on wire (2736 bits), 342 bytes captured (2736
Ethernet II, Src: PCSSystemtec_7a:9e:2b (08:00:27:7a:9e:2b), Dst
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
User Datagram Protocol, Src Port: 68, Dst Port: 67
Dynamic Host Configuration Protocol (Discover)
0000 ff ff ff ff ff ff 08 00 27 7a 9e 2b 08 00 45 10
0010 01 48 00 00 00 00 80 11 39 96 00 00 00 00 ff ff
0020 ff ff 00 44 00 43 01 34 96 5a 01 01 06 00 40 27
0030 31 e4 f5 4a 00 00 00 00 00 00 00 00 00 00 00 00
0040 00 00 00 00 00 00 08 00 27 7a 9e 2b 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
wireshark_enp0s363WQW3.pcapng
Packets: 16 · Displayed: 16 (100.0%) · Dropped: 0 (0.0%)
Profile: Default


*enp0s3
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
arp
No. Time Source Destination Protocol Length Info
14 74.803363902 PCSSystemtec_9e:59:… PCSSystemtec_7a:9e:… ARP 42 Who has 10.10.10.1? Tell 10.10.10.50
15 74.804782800 PCSSystemtec_7a:9e:… PCSSystemtec_9e:59:… ARP 60 10.10.10.1 is at 08:00:27:7a:9e:2b
Frame 14: 42 bytes on wire (336 bits), 42 bytes captured (336 bi
Ethernet II, Src: PCSSystemtec_9e:59:4f (08:00:27:9e:59:4f), Dst
Address Resolution Protocol (request)
0000 08 00 27 7a 9e 2b 08 00 27 9e 59 4f 08 06 00 01
0010 08 00 06 04 00 01 08 00 27 9e 59 4f 0a 0a 0a 32
0020 00 00 00 00 00 00 0a 0a 0a 01
Address Resolution Protocol: Protocol
Packets: 16 · Displayed: 2 (12.5%) · Dropped: 0 (0.0%)
Profile: Default


*enp0s3
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Length Info
6 69.323455133 10.10.10.50 10.10.10.1 ICMP 98 Echo (ping) request id=0x3d82, seq=1/256, ttl=64 (repl
7 69.324011853 10.10.10.1 10.10.10.50 ICMP 98 Echo (ping) reply id=0x3d82, seq=1/256, ttl=64 (requ
8 70.392306722 10.10.10.50 10.10.10.1 ICMP 98 Echo (ping) request id=0x3d82, seq=2/512, ttl=64 (repl
9 70.394140608 10.10.10.1 10.10.10.50 ICMP 98 Echo (ping) reply id=0x3d82, seq=2/512, ttl=64 (requ
10 71.460285903 10.10.10.50 10.10.10.1 ICMP 98 Echo (ping) request id=0x3d82, seq=3/768, ttl=64 (repl
11 71.464122887 10.10.10.1 10.10.10.50 ICMP 98 Echo (ping) reply id=0x3d82, seq=3/768, ttl=64 (requ
12 72.463135200 10.10.10.50 10.10.10.1 ICMP 98 Echo (ping) request id=0x3d82, seq=4/1024, ttl=64 (rep
13 72.465316443 10.10.10.1 10.10.10.50 ICMP 98 Echo (ping) reply id=0x3d82, seq=4/1024, ttl=64 (req
Frame 13: 98 bytes on wire (784 bits), 98 bytes captured (784 bi
Ethernet II, Src: PCSSystemtec_7a:9e:2b (08:00:27:7a:9e:2b), Dst
Internet Protocol Version 4, Src: 10.10.10.1, Dst: 10.10.10.50
Internet Control Message Protocol
0000 08 00 27 9e 59 4f 08 00 27 7a 9e 2b 08 00 45 00
0010 00 54 2e be 00 00 40 01 23 a5 0a 0a 0a 01 0a 0a
0020 0a 32 00 00 7b b8 3d 82 00 04 0b 0c c5 6a 00 00
0030 00 00 ac 77 0b 00 00 00 00 00 10 11 12 13 14 15
0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25
0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35
0060 36 37
Internet Control Message Protocol: Protocol
Packets: 16 · Displayed: 8 (50.0%) · Dropped: 0 (0.0%)
Profile: Default



A short paragraph explaining why icdfa-nslab-client-v1 uses 10.10.10.1 as its default gateway.

The icdfa-nslab-client-v1 uses 10.10.10.1 as its default gateway because this is the LAN address of the OPNsense firewall. The client is connected to the ICDFA-LAN network, which uses the 10.10.10.0/24 subnet. Therefore, any traffic destined for networks outside the local subnet, including the Internet, is forwarded to the OPNsense LAN interface at 10.10.10.1, which then routes the traffic through its WAN interface.

What is the difference between the OPNsense WAN and LAN interfaces?

The WAN interface connects OPNsense to the external network or Internet through VirtualBox NAT, while the LAN interface connects OPNsense to the internal ICDFA LAN network and provides services such as routing and DHCP to the Ubuntu client.

Why must both internal adapters use the same VirtualBox network name?

Both adapters must use the same VirtualBox network name so that they belong to the same isolated virtual Ethernet network. This allows the Ubuntu client and the OPNsense LAN interface to communicate directly.

What information does the default route provide to Ubuntu?

The default route tells Ubuntu where to send packets destined for networks that are not on its local network. In this setup, the default gateway is 10.10.10.1, the OPNsense LAN interface.

Which packet exchange allows Ubuntu to learn the firewall MAC address?

ARP, Address Resolution Protocol, allows Ubuntu to discover the MAC address associated with 10.10.10.1. Ubuntu sends an ARP request asking which device owns that IP, and OPNsense responds with its MAC address.

5. Why does a successful ping to 1.1.1.1 not automatically prove that DNS is working?

A ping to 1.1.1.1 uses an IP address directly, so DNS is not involved. DNS must be tested separately using a hostname such as opnsense.org to confirm that Ubuntu can translate domain names into IP addresses.